



Digital Transformations for
Supporting Next-Generation Labour

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of the European Union



MetaLab

MacPaw AI Lab in VRChat for the NEXT-Study Metaverse

Nomination I — Metaverse representation of the university

Team MetaLab

Igor Sikorsky Kyiv Polytechnic Institute

Team members

Sviatoslav Pavlenko, Denys Iliencko, Dmytro Poshytyniuk,
Mark Mamenko, Kateryna Shozda

nextstudy.eu

Sviatoslav Pavlenko

Igor Sikorsky KPI

Role in the project

Team Lead. Coordination, task split, deadlines and communication with the consultant.

Denys Ilienکو

Igor Sikorsky KPI

Role in the project

3D Artist. Room and equipment modelling, retopology, mesh prep for texturing.

Dmytro Poshytyniuk

Igor Sikorsky KPI

Role in the project

Look Development. PBR textures and Unity materials for realistic surfaces.

Mark Mamenko

Igor Sikorsky KPI

Role in the project

UdonSharp Developer. Interactive terminals, object state sync across players.

Kateryna Shozda

Igor Sikorsky KPI

Role in the project

QA / VR Tester. In-headset checks, FPS, collisions and network stability.

How we work

Five people. Modelling, materials, UdonSharp and Unity assembly run in parallel and sync through Git, with regular tests in a headset.

ANNOTATION

Digital twin of our lab in VRChat

MacPaw AI Lab · NEXT-Study Metaverse

For the contest we published a multiplayer VRChat copy of our MacPaw AI Lab: seminar rows, a lounge, boards and workstations in one hall, so a group shares a room instead of a video grid. Built in Unity 2022.3.22f1 and VRChat SDK3, with UdonSharp, Blender retopology, Substance PBR and baked light.

PROJECT GOALS

- Visual accuracy: real proportions of furniture and gear
- Functional space for SE and networking practice
- Extensible layout for later AI / simulators
- Stable FPS via retopo, PBR and light baking
- Multiplayer VRChat world for NEXT-Study

Selected Location

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MacPaw AI Lab

The photos are our laboratory only: overview, lounge and seminar rows. No stock offices and no generated interiors.



The space in VRChat

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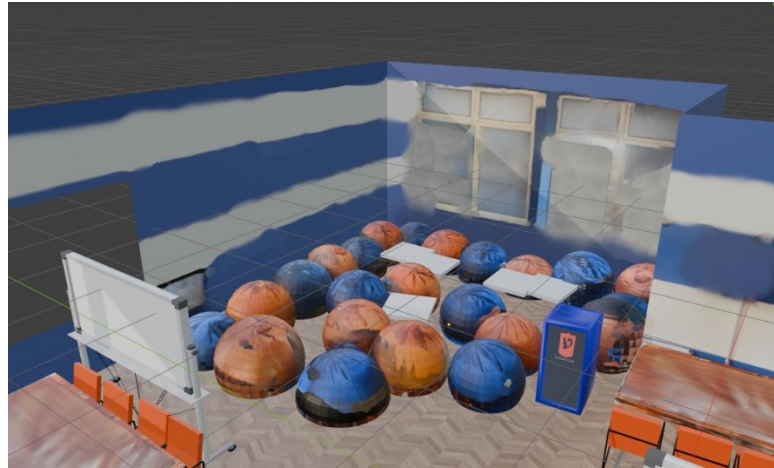


Classroom layout



Retopologised mesh, baked light instead of realtime lamps, colliders on the floor, tables and walls.

Lounge and boards



Lounge and boards. The wall logo is a raster from our lab, not a generated picture.

Workstations



Workstations plus UdonSharp sync. Space is left for the CodeSensei mentor.

Tools and software

Unity (scene, collisions, light baking) · VRChat SDK3 / Udon · UdonSharp (C#) · Blender · Adobe Substance 3D Painter (PBR) · Git.

Development process

1) Block out MacPaw AI Lab. 2) Model and retopo furniture and equipment. 3) Paint PBR materials. 4) Bake lighting and collisions in Unity. 5) Add UdonSharp interaction and publish the world.

VRChat integration

SDK3 Worlds: spawns, audio zones, synced object state. Published world wrld_b1c73436-f022-4f98-9172-671f9f0da989. Prefer PC / PCVR.

Difficulties and limits

VRChat caps polygons and materials, so meshes were retopologised. Light is baked. Only Udon can sync props. Quest needs a separate pass. CodeSensei calls need Allow Untrusted URLs.

Integration into NEXT-Study Metaverse

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Educational use

A class meets in this hall for a remote practical, a hackathon or an applicant tour, with the spatial feel of the real lab.

User experience

Spawn, walk the aisles, sit at a workstation, use boards and equipment, see classmates in the same room. Presence replaces a flat video call.

Further development

Keep the CodeSensei terminal in the hall, add small network simulators over OSC or HTTP, split gear into prefabs, and make a Quest pass.

Demo and Project Materials

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Project demo

https://vrchat.com/home/world/wrld_b1c73436-f022-4f98-9172-671f9f0da989/info

Project materials

GitHub: github.com/mamenkomarkus-creator/MetaLab-NEXT

Docs: presentation-materials/documentation/DOCUMENTATION.md

World ID: wrld_b1c73436-f022-4f98-9172-671f9f0da989

Additional resources

VRChat: hello.vrchat.com NEXT: nextstudy.eu



GitHub



VRChat



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Thank you!
Questions?

